

1. (a) Give an example of linear regression model with single regressor (X_1) and omitted variable (X_2) that has concern of OVB.
(b) Explain the meaning of dummy variable trap (DVT). Give an example of DVT.
2. The following equation describes the housing price in a community in terms of amount of pollution (nox for nitrous oxide) and the number of rooms in houses in the community ($rooms$):

$$\begin{aligned} price_i &= \beta_0 + \beta_1 nox_i + \beta_2 rooms_i + e_i, i = 1, \dots, n; \\ E[e_i | nox_i, rooms_i] &= 0. \end{aligned}$$

- (a) Explain the two conditions of omitted variable bias.
 - (b) What are the probable signs of β_1 and β_2 ?
 - (c) Suppose that nox_i and $rooms_i$ are negatively correlated. Does the simple regression of $price_i$ on nox_i produce an upward or a downward biased estimator of β_1 ? Explain.
3. This question is based on the 2015 Current Population Survey in US. The dataset consists of information on 7178 full-time, full-year workers. The highest educational achievement for each worker was either a high school diploma or a bachelor's degree. The workers' ages ranged from 25 to 34 years. The dataset also contains information on the region of the country where the person lived, marital status, and number of children. Let AHE = average hourly earnings, $College$ = binary variable (1 if college, 0 if high school), $Female$ = binary variable (1 if female, 0 if male), Age = age (in years), $Northeast$ = binary variable (1 if Region = Northeast, 0 otherwise), $Midwest$ = binary variable (1 if Region = Midwest, 0 otherwise), $South$ = binary variable (1 if Region = South, 0 otherwise), $West$ = binary variable (1 if Region = West, 0 otherwise).
 - (a) Using the regression results in column (1): Do workers with college degrees earn more, on average, than workers with only high school diplomas? How much more?
 - (b) Using the regression results in column (2): Sally is a 29-year-old female college graduate. Besty is a 34-year-old female college graduate. Predict Sally's and Besty's earnings.
 - (c) Using the regression results in column (3):
 - i. Why is the regressor $West$ omitted from the regression? What would happen if it were included?

- ii. Juanita is a 28-year-old female college graduate from the South. Jennifer is a 25-year-old female college graduate from the Midwest. Calculate the predicted difference in earnings between the two.

Results of Regressions of Average Hourly Earnings on Sex and Education Binary Variables and Other Characteristics, Using 2015 Data from the Current Population Survey			
Dependent variable: average hourly earnings (AHE).			
Regressor	(1)	(2)	(3)
College (X_1)	10.47	10.44	10.42
Female (X_2)	-4.69	-4.56	-4.57
Age (X_3)		0.61	0.61
Northeast (X_4)			0.74
Midwest (X_5)			-1.54
South (X_6)			-0.44
Intercept	18.15	0.11	0.33

4. Empirical exercise: Use the dataset **birthweight_smoking.dta** uploaded in NTULearn to answer the following questions (STATA is available in the computer labs at B1 of the SHHK building, but you are welcome to use other software if you prefer).
- Regress “birthweight” (birth weight of infant, in grams) on “smoker” (indicator variable equal to one if the mother smoked during pregnancy, and zero otherwise). What is the estimated effect of smoking on birth weight?
 - Regress “birthweight” on “smoker” and “nprevisit” (total number of prenatal care visits). How is the estimated effect of smoking different from the result in (a)? Show the sign of the potential omitted variable bias of the regression with the single regressor “smoker”. (Note: In STATA, we can check the correlation between two variables X and Y using the following command: **correlate X Y**)
 - Jane smoked during her pregnancy, and had 8 prenatal care visits. Use the regression to predict the birth weight of Jane’s child.